FOURIER TRANSFORM ON $\mathbb{R}$

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ABSTRACT. These notes develop the Fourier transform on $\mathbb{R}$ from translation symmetry, establish its basic algebraic and analytic properties, construct Gaussian approximate identities, and prove inversion for $f, \widehat{f} \in L^1(\mathbb{R})$. The Gaussian, interval, and triangular examples precede concise statements of Plancherel and Hausdorff–Young.

CONTENTS

Part 1. Fourier analysis on the real line

1. MOTIVATION, NORMALIZATION, AND ANALYTIC PRELIMINARIES

Translation symmetry determines the Fourier kernel, while convolution and the standard integration theorems supply the analytic framework used throughout [6, 1, 4]. We first fix the normalization and the few tools needed later.

Remark 1.1 (Motivation). Functions on $\mathbb{R}$ are analyzed through the oscillations determined by translation symmetry. The simultaneous eigenfunctions of translation are precisely the continuous characters of $(\mathbb{R}, +)$.

For $t \in \mathbb{R}$, let $T_t f(x) := f(x-t)$. The translation operators satisfy

$$T_s \circ T_t = T_{s+t}, \qquad T_0 = \mathrm{id}, \qquad T_t^{-1} = T_{-t}.$$

A nonzero simultaneous eigenfunction obeys

$$T_t \phi = \lambda(t)\phi \implies \lambda(t+s) = \lambda(t)\lambda(s), \qquad \lambda(0) = 1.$$

Continuity gives $\lambda(t) = \mathrm{e}^{ct}$ for some $c \in \mathbb{C}$, while unitarity on $L^2(\mathbb{R})$ forces $c = -2\pi \mathrm{i}\xi$ with $\xi \in \mathbb{R}$. Thus

$$\chi_\xi(x) := \mathrm{e}^{2\pi \mathrm{i}\xi x}, \qquad T_t \chi_\xi = \mathrm{e}^{-2\pi \mathrm{i} t\xi} \chi_\xi.$$

Definition 1.2 (Fourier transform, [6])**.** For $f \in L^1(\mathbb{R})$, the *Fourier transform* of f is

$$\widehat{f}(\xi) := \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} x\xi}\, dx, \qquad \xi \in \mathbb{R}.$$

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Remark 1.3 (Normalization). The factor 2π places the inverse kernel at $\mathrm{e}^{2\pi\mathrm{i}x\xi}$ and makes

$$g(x) := \mathrm{e}^{-\pi x^2} \qquad \Longrightarrow \qquad \widehat{g}(\xi) = \mathrm{e}^{-\pi\xi^2} = g(\xi).$$

Remark 1.4 (Structural meaning). The kernel in Definition 1.2 is forced by translation symmetry:

$$T_t f \xrightarrow{\mathscr{F}} \mathrm{e}^{-2\pi\mathrm{i}t\xi}\widehat{f}(\xi).$$

For $1 \le p < \infty$, write

$$\|f\|_p := \left(\int_{\mathbb{R}} |f|^p\right)^{1/p}, \qquad \|f\|_\infty := \operatorname{ess\,sup}_{x\in\mathbb{R}}|f(x)|.$$

The Schwartz class is

$$\mathscr{S}(\mathbb{R}) := \left\{ f \in C^\infty(\mathbb{R}) : \sup_{x\in\mathbb{R}} |x^m f^{(n)}(x)| < \infty \quad (m,n \in \mathbb{N}\cup\{0\}) \right\}.$$

Definition 1.5 (Convolution)**.** For $f,g \in L^1(\mathbb{R})$, their *convolution* is

$$(f*g)(x) := \int_{\mathbb{R}} f(x-y)g(y)\,dy, \qquad \|f*g\|_1 \le \|f\|_1\|g\|_1.$$

Lemma 1.6 (Integration tools, [1, 4])**.** *Let $h : \mathbb{R}^2 \to \mathbb{C}$ and $u_n, u : \mathbb{R} \to \mathbb{C}$ be measurable. Then*

$$\int_{\mathbb{R}^2} |h(x,y)|\,dx\,dy < \infty \implies \int_{\mathbb{R}}\int_{\mathbb{R}} h(x,y)\,dx\,dy = \int_{\mathbb{R}}\int_{\mathbb{R}} h(x,y)\,dy\,dx,$$

$$u_n \to u \text{ a.e.}, \qquad |u_n| \le H \in L^1(\mathbb{R}) \implies \int_{\mathbb{R}} u_n \to \int_{\mathbb{R}} u.$$

Remark 1.7 (Analytic tools). Lemma 1.6 governs convolution and Gaussian damping, while Lebesgue differentiation governs pointwise recovery.

2. Transform identities and decay

The transform converts elementary operations on f into explicit operations on $\widehat{f}$, while the Riemann–Lebesgue lemma records the first general decay principle [6, 7, 2]. These identities form the computational core of the notes.

Proposition 2.1 (Boundedness and continuity)**.** *If $f \in L^1(\mathbb{R})$, then*

$$\|\widehat{f}\|_\infty \le \|f\|_1, \qquad \widehat{f} \in C(\mathbb{R}) \quad \textit{uniformly continuously}.$$

Proof. Fix $\xi, h \in \mathbb{R}$ and estimate directly.

$$|\widehat{f}(\xi)| \le \int_{\mathbb{R}} |f(x)|\,dx = \|f\|_1.$$

$$\sup_{\xi\in\mathbb{R}} |\widehat{f}(\xi+h) - \widehat{f}(\xi)| \le \int_{\mathbb{R}} |f(x)|\,|\mathrm{e}^{-2\pi\mathrm{i}xh} - 1|\,dx.$$

$$|\mathrm{e}^{-2\pi\mathrm{i}xh} - 1| \to 0, \qquad |f(x)|\,|\mathrm{e}^{-2\pi\mathrm{i}xh} - 1| \le 2|f(x)| \overset{1.6}{\Longrightarrow} \sup_{\xi\in\mathbb{R}} |\widehat{f}(\xi+h) - \widehat{f}(\xi)| \to 0.$$

This proves both assertions. □

Proposition 2.2 (Linearity, conjugation, and reflection)**.** *For $a,b \in \mathbb{C}$, $f,g \in L^1(\mathbb{R})$, and $f^\sharp(x) := f(-x)$,*

$$\widehat{af+bg} = a\widehat{f} + b\widehat{g}, \qquad \widehat{\overline{f}}(\xi) = \overline{\widehat{f}(-\xi)}, \qquad \widehat{f^\sharp}(\xi) = \widehat{f}(-\xi).$$

Proposition 2.3 (Transform rules). *Let $t, \xi_0 \in \mathbb{R}$ and $a \in \mathbb{R}\setminus\{0\}$. For every admissible f,*

$$\widehat{T_t f}(\xi) = \mathrm{e}^{-2\pi\mathrm{i}t\xi}\widehat{f}(\xi), \qquad \widehat{\mathrm{e}^{2\pi\mathrm{i}\xi_0(\cdot)}f}(\xi) = \widehat{f}(\xi-\xi_0), \qquad \widehat{f(a\,\cdot)}(\xi) = \frac{1}{|a|}\widehat{f}\left(\frac{\xi}{a}\right).$$

If $f \in W^{1,1}(\mathbb{R})$, or if $f \in L^1(\mathbb{R})$ and $xf(x) \in L^1(\mathbb{R})$, respectively, then

$$\widehat{f'}(\xi) = 2\pi\mathrm{i}\xi\widehat{f}(\xi), \qquad \widehat{xf}(\xi) = -\frac{1}{2\pi\mathrm{i}}\frac{d}{d\xi}\widehat{f}(\xi).$$

Proof. Each rule follows from substitution, integration by parts, or differentiation under the integral.

$$\int_{\mathbb{R}} f(x-t)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx = \mathrm{e}^{-2\pi\mathrm{i}t\xi}\widehat{f}(\xi), \qquad \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi\mathrm{i}x(\xi-\xi_0)}\,dx = \widehat{f}(\xi-\xi_0).$$

$$\int_{\mathbb{R}} f(ax)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx = \frac{1}{|a|}\int_{\mathbb{R}} f(u)\mathrm{e}^{-2\pi\mathrm{i}u\xi/a}\,du = \frac{1}{|a|}\widehat{f}\left(\frac{\xi}{a}\right).$$

$$\int_{\mathbb{R}} f'(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx = 2\pi\mathrm{i}\xi\int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx = 2\pi\mathrm{i}\xi\widehat{f}(\xi).$$

$$\frac{d}{d\xi}\widehat{f}(\xi) = -2\pi\mathrm{i}\int_{\mathbb{R}} xf(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx = -2\pi\mathrm{i}\widehat{xf}(\xi).$$

This proves the transform rules. □

Proposition 2.4 (Convolution theorem). *For $f, g \in L^1(\mathbb{R})$,*

$$\widehat{f*g} = \widehat{f}\cdot\widehat{g}.$$

Proof. Apply Fubini and set $u = x-y$.

$$\int_{\mathbb{R}^2} |f(x-y)g(y)|\,dy\,dx = \|f\|_1\|g\|_1 < \infty.$$

$$\widehat{f*g}(\xi) = \int_{\mathbb{R}^2} f(x-y)g(y)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dy\,dx.$$

$$\widehat{f*g}(\xi) \overset{1.6}{\Longrightarrow} \left(\int_{\mathbb{R}} f(u)\mathrm{e}^{-2\pi\mathrm{i}u\xi}\,du\right)\cdot\left(\int_{\mathbb{R}} g(y)\mathrm{e}^{-2\pi\mathrm{i}y\xi}\,dy\right) = \widehat{f}(\xi)\widehat{g}(\xi).$$

This proves the convolution identity. □

Lemma 2.5 (Riemann–Lebesgue, [6, 2]). *For $f \in L^1(\mathbb{R})$,*

$$\widehat{f}(\xi) \longrightarrow 0 \qquad (|\xi| \to \infty).$$

Proof. Approximate f in L^1 by $g \in C_c^1(\mathbb{R})$.

$$|\widehat{f}(\xi)| \le \|f-g\|_1 + |\widehat{g}(\xi)| \le \|f-g\|_1 + \frac{\|g'\|_1}{2\pi|\xi|} \qquad (\xi \neq 0).$$

$$\varepsilon > 0, \qquad \|f-g\|_1 < \frac{\varepsilon}{2}, \qquad |\xi| > \frac{\|g'\|_1}{\pi\varepsilon} \Longrightarrow |\widehat{f}(\xi)| < \varepsilon.$$

This proves the asserted decay. □

Remark 2.6 (Time and frequency). The preceding identities compress to

$$T_t f \leftrightarrow \mathrm{e}^{-2\pi\mathrm{i}t\xi}\widehat{f}, \qquad f*g \leftrightarrow \widehat{f}\cdot\widehat{g}, \qquad f' \leftrightarrow 2\pi\mathrm{i}\xi\widehat{f}.$$

3. Gaussian regularization and Fourier inversion

Gaussian kernels connect spatial approximation with frequency damping and give a direct route to inversion [1, 4, 8]. We first establish concentration in physical space and then remove the corresponding Gaussian multiplier.

Let $g(x) := \mathrm{e}^{-\pi x^2}$ and, for $\varepsilon > 0$, set

$$\phi_\varepsilon(x) := \frac{1}{\varepsilon} g\Big(\frac{x}{\varepsilon}\Big) = \frac{1}{\varepsilon}\mathrm{e}^{-\pi(x/\varepsilon)^2}, \qquad \int_{\mathbb{R}} \phi_\varepsilon = 1, \qquad \widehat{\phi_\varepsilon}(\xi) = \mathrm{e}^{-\pi\varepsilon^2\xi^2}.$$

The last identity follows from Example 4.1 and Proposition 2.3.

Proposition 3.1 (Gaussian approximate identity, [1, 8])**.** *For $f \in L^1(\mathbb{R})$,*

$$\|f * \phi_\varepsilon - f\|_1 \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

At every Lebesgue point x of f,

$$(f * \phi_\varepsilon)(x) \longrightarrow f(x) \qquad (\varepsilon \downarrow 0).$$

Remark 3.2 (Lebesgue points, [1, 4]). Almost every $x \in \mathbb{R}$ is a Lebesgue point of every $f \in L^1(\mathbb{R})$.

Remark 3.3 (Local averaging). The kernel ϕ_ε is positive, has mass one, and concentrates at the origin:

$$(f * \phi_\varepsilon)(x) = \int_{\mathbb{R}} f(x - \varepsilon u) g(u)\, du \xrightarrow{\varepsilon\downarrow 0} f(x).$$

The same Gaussian damping makes the inverse integral absolutely convergent before the damping is removed, yielding inversion without first assuming it on the Schwartz class.

Remark 3.4 (Proof strategy). The argument has three steps:

$$\text{regularize} \xrightarrow{\text{Fubini}} \text{invert the Gaussian-damped transform} \xrightarrow{\varepsilon\downarrow 0} \text{recover } f.$$

$$f * \phi_\varepsilon = \mathscr{F}^{-1}\Big(\widehat{f}\mathrm{e}^{-\pi\varepsilon^2(\cdot)^2}\Big), \qquad \varepsilon \downarrow 0 \implies f * \phi_\varepsilon \to f.$$

Lemma 3.5 (Gaussian regularization)**.** *For $f \in L^1(\mathbb{R})$, $x \in \mathbb{R}$, and $\varepsilon > 0$,*

$$(f * \phi_\varepsilon)(x) = \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\, d\xi.$$

Proof. Expand $\widehat{f}$ and apply Fubini to the absolutely integrable kernel.

$$\int_{\mathbb{R}^2} |f(u)|\mathrm{e}^{-\pi\varepsilon^2\xi^2}\, du\, d\xi = \frac{\|f\|_1}{\varepsilon} < \infty.$$

$$\int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\, d\xi = \int_{\mathbb{R}} f(u)\left(\int_{\mathbb{R}} \mathrm{e}^{-\pi\varepsilon^2\xi^2}\mathrm{e}^{2\pi \mathrm{i}(x-u)\xi}\, d\xi\right) du.$$

$$\int_{\mathbb{R}} \mathrm{e}^{-\pi\varepsilon^2\xi^2}\mathrm{e}^{2\pi \mathrm{i}(x-u)\xi}\, d\xi = \phi_\varepsilon(x-u) \implies \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\, d\xi = (f * \phi_\varepsilon)(x).$$

This proves the regularization identity. □

Proposition 3.6 (Inversion on the Schwartz class)**.** *For $f \in \mathscr{S}(\mathbb{R})$ and every $x \in \mathbb{R}$,*

$$f(x) = \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi \mathrm{i} x\xi}\, d\xi.$$

Proof. Apply Lemma 3.5 and let $\varepsilon \downarrow 0$.

$$(f*\phi_\varepsilon)(x) = \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\,d\xi.$$

$$f\in\mathscr{S}(\mathbb{R}) \overset{2.3}{\Longrightarrow} \widehat{f}\in\mathscr{S}(\mathbb{R})\subset L^1(\mathbb{R}), \qquad 0<\mathrm{e}^{-\pi\varepsilon^2\xi^2}\le 1.$$

$$\int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\,d\xi \xrightarrow[\varepsilon\downarrow 0]{1.6} \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\,d\xi.$$

$$\sup_{x\in\mathbb{R}}|(f*\phi_\varepsilon)(x)-f(x)| \le \int_{\mathbb{R}} g(u)\sup_{x\in\mathbb{R}}|f(x-\varepsilon u)-f(x)|\,du \longrightarrow 0.$$

This proves inversion on $\mathscr{S}(\mathbb{R})$. □

Theorem 3.7 (Fourier inversion, [6, 1, 4])**.** *If $f\in L^1(\mathbb{R})$ and $\widehat{f}\in L^1(\mathbb{R})$, then*

$$f(x) = \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\,d\xi \qquad \text{for almost every } x\in\mathbb{R}.$$

Equivalently, $\mathscr{F}^{-1}(\widehat{f}) = f$ almost everywhere.

Proof. Fix a Lebesgue point x of f and use Gaussian regularization.

$$(f*\phi_\varepsilon)(x) \overset{3.5}{\Longrightarrow} \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\,d\xi.$$

$$\widehat{f}\in L^1(\mathbb{R}), \qquad |\widehat{f}(\xi)\mathrm{e}^{-\pi\varepsilon^2\xi^2}|\le|\widehat{f}(\xi)| \overset{1.6}{\Longrightarrow} \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\mathrm{e}^{-\pi\varepsilon^2\xi^2}\,d\xi \longrightarrow \int_{\mathbb{R}} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\,d\xi.$$

$$x \text{ a Lebesgue point} \overset{3.1}{\Longrightarrow} (f*\phi_\varepsilon)(x)\longrightarrow f(x), \qquad \text{a.e. } x \text{ is a Lebesgue point.}$$

This proves the inversion formula almost everywhere. □

Remark 3.8 (Pointwise interpretation, [7]). If f is continuous at x, Proposition 3.1 gives the formula of Theorem 3.7 at x; if f satisfies the local Dirichlet hypotheses there, symmetric Fourier truncation gives

$$\lim_{A\to\infty}\int_{-A}^{A} \widehat{f}(\xi)\mathrm{e}^{2\pi\mathrm{i}x\xi}\,d\xi = \frac{f(x-)+f(x+)}{2}.$$

Moreover, the inverse integral is continuous whenever $\widehat{f}\in L^1(\mathbb{R})$.

4. Examples and extensions

The normalization of Definition 1.2 makes the Gaussian, interval, and triangular transforms explicit; completion and interpolation then extend the theory to L^2 and intermediate exponents [6, 7, 2]. The examples and extensions below display both sides of this passage.

Example 4.1 (Gaussian self-duality)**.** Let $g(x) := \mathrm{e}^{-\pi x^2}$ and $G := \widehat{g}$. Differentiation under the integral and integration by parts give

$$G'(\xi) = -2\pi\mathrm{i}\int_{\mathbb{R}} xg(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx, \qquad g'(x) = -2\pi xg(x).$$

$$\int_{\mathbb{R}} g'(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,dx = 2\pi\mathrm{i}\xi G(\xi) \implies G'(\xi) = -2\pi\xi G(\xi).$$

$$G(0) = \int_{\mathbb{R}} \mathrm{e}^{-\pi x^2}\,dx = 1 \implies G(\xi) = \mathrm{e}^{-\pi\xi^2} = g(\xi).$$

Example 4.2 (Interval to sinc)**.** For $f := \mathbf{1}_{[-1,1]}$,

$$\widehat{f}(\xi) = \int_{-1}^{1} \mathrm{e}^{-2\pi \mathrm{i} x\xi}\, dx = \begin{cases} \dfrac{\sin(2\pi\xi)}{\pi\xi}, & \xi \neq 0, \\ 2, & \xi = 0. \end{cases}$$

The $O(|\xi|^{-1})$ envelope reflects the two jump discontinuities.

Example 4.3 (Triangle by convolution)**.** Let $f := \mathbf{1}_{[-1,1]}$ and $\tau := f * f$. Proposition 2.4 gives

$$\tau(x) = \begin{cases} 2-|x|, & |x| \leq 2, \\ 0, & |x| > 2, \end{cases} \qquad \widehat{\tau}(\xi) = \widehat{f}(\xi)^2 = \left(\frac{\sin(2\pi\xi)}{\pi\xi}\right)^2.$$

Thus one convolution raises the decay envelope from $O(|\xi|^{-1})$ to $O(|\xi|^{-2})$.

Beyond the model computations, the L^1 theory extends to L^2 by completion and to intermediate exponents by interpolation [7, 2].

Theorem 4.4 (Plancherel, [6, 1, 7])**.** *The Fourier transform extends uniquely to a unitary operator on* $L^2(\mathbb{R})$*:*

$$\|f\|_2 = \|\widehat{f}\|_2, \qquad \langle f, g\rangle = \langle \widehat{f}, \widehat{g}\rangle, \qquad \mathscr{F}^{-1} \circ \mathscr{F} = \mathrm{id}_{L^2(\mathbb{R})}.$$

Remark 4.5 (Proof mechanism). Prove the identity on $\mathscr{S}(\mathbb{R})$, use density in $L^2(\mathbb{R})$, and extend by continuity:

$$\mathscr{S}(\mathbb{R}) \subset L^2(\mathbb{R}) \text{ dense}, \qquad \|\widehat{f}\|_2 = \|f\|_2 \implies \mathscr{F} : L^2(\mathbb{R}) \xrightarrow{\cong} L^2(\mathbb{R}).$$

Theorem 4.6 (Hausdorff–Young, [2, 7])**.** *If* $1 \leq p \leq 2$*,* $p^{-1} + q^{-1} = 1$*, and* $f \in L^p(\mathbb{R})$*, then*

$$\widehat{f} \in L^q(\mathbb{R}), \qquad \|\widehat{f}\|_q \leq \|f\|_p.$$

Remark 4.7 (Interpolation). The Riesz–Thorin theorem interpolates the endpoints

$$\mathscr{F} : L^1(\mathbb{R}) \to L^\infty(\mathbb{R}), \qquad \mathscr{F} : L^2(\mathbb{R}) \to L^2(\mathbb{R}).$$

Fourier-series analogues on $\mathbb{T}$ appear in [3]; complex-analytic boundary connections appear in [5].

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